

Day 10 – File Permissions & File Operations Challenge

Task 1: Create Files

1. Create empty file devops.txt

touch devops.txt

```
ubuntu@ip-172-31-44-9:~$ touch devops.txt
ubuntu@ip-172-31-44-9:~$
```

Verify:

ls -l

```
ubuntu@ip-172-31-44-9:~$ ls -l
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 May 31 14:46 devops.txt
ubuntu@ip-172-31-44-9:~$
```

2. Create notes.txt with content

Using echo:

echo "Hello Dosto." > notes.txt

```
ubuntu@ip-172-31-44-9:~$ echo "hello dosto">notes.txt
ubuntu@ip-172-31-44-9:~$ ls
cat  devops.txt  notes.txt
ubuntu@ip-172-31-44-9:~$ cat notes.txt
hello dosto
ubuntu@ip-172-31-44-9:~$
```

Check content:

cat notes.txt

3. Create script.sh

```
ubuntu@ip-172-31-44-9:~$ vim script.sh
ubuntu@ip-172-31-44-9:~$ cat script.sh
Hello Devops
ubuntu@ip-172-31-44-9:~$
```

4. Verify files and permissions

ls -l

```
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-rw-rw-r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Task 2: Read Files

1. Read notes.txt

cat notes.txt

```
ubuntu@ip-172-31-44-9:~$ cat notes.txt
hello dosto
ubuntu@ip-172-31-44-9:~$
```

2. Open script.sh in read-only mode

```
vim -R script.sh
```

[illegible]

Exit:

:q

3. Display first 5 lines of /etc/passwd

```
head -n 5 /etc/passwd
```

4. Display last 5 lines of /etc/passwd

```
tail -n 5 /etc/passwd
```

```
ubuntu@ip-172-31-44-9:~$ head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
ubuntu@ip-172-31-44-9:~$ tail -5 /etc/passwd
landscape:x:103:106:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:984:984:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:983:983:User for polkitd:/usr/sbin/nologin
ec2-instance-connect:x:104:65534:./nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
ubuntu@ip-172-31-44-9:~$
```

Task 3: Understand Permissions

Run:

```
ls -l
```

```
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-rw-rw-r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 13 May 31 14:52 script.sh
```

Understanding

Owner Group Others

-rw- rw- r--

Meaning:

- Owner can read and write
- Group can read and write
- Others can only read
- Nobody can execute
- - denotes file

Permission Numbers

r = 4

w = 2

x = 1

Examples:

rw- = 7

rw- = 6

r-- = 4

--- = 0

Task 4: Modify Permissions

1. Make script.sh executable

chmod +x script.sh

Check:

ls -l script.sh

```
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-rw-rw-r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$ chmod +x script.sh
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-rw-rw-r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Run:

[./script.sh](#)

```
ubuntu@ip-172-31-44-9:~$ chmod +x script.sh
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-rw-rw-r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$ ./script.sh
./script.sh: line 1: Hello: command not found
ubuntu@ip-172-31-44-9:~$
```

2. Set devops.txt to read-only

chmod 444 devops.txt

-r--r--r--

Meaning:

- Everyone can read

- Nobody can write
- Nobody can execute

```
ubuntu@ip-172-31-44-9:~$ chmod 444 devops.txt
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-r--r--r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

3. Set notes.txt permission to 640

`chmod 640 notes.txt`

Check:

```
ubuntu@ip-172-31-44-9:~$ chmod 640 notes.txt
ubuntu@ip-172-31-44-9:~$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-r--r--r-- 1 ubuntu ubuntu  0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

`-rw-r-----`

Meaning:

- Owner → read/write
- Group → read only
- Others → no access

4. Create directory project/ with 755 permission

`mkdir project`

`chmod 755 project`

```
ubuntu@ip-172-31-44-9:~$ mkdir project
ubuntu@ip-172-31-44-9:~$ ls -l
total 16
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-r--r--r-- 1 ubuntu ubuntu 0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 May 31 15:06 project
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$ chmod 755 project
ubuntu@ip-172-31-44-9:~$ ls -l
total 16
-rw-rw-r-- 1 ubuntu ubuntu 22 May 31 14:48 cat
-r--r--r-- 1 ubuntu ubuntu 0 May 31 14:46 devops.txt
-rw-r----- 1 ubuntu ubuntu 12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 May 31 15:06 project
-rwxrwxr-x 1 ubuntu ubuntu 13 May 31 14:52 script.sh
ubuntu@ip-172-31-44-9:~$
```

Check:

ls -ld project

Expected:

drwxr-xr-x

Meaning:

- Owner → full access
- Group → read/execute
- Others → read/execute

Task 5: Test Permissions

1. Try writing to read-only file

```
echo "test" >> devops.txt
```

Permission denied


```
ubuntu@ip-172-31-44-9:~$ echo "hello parthavi">>devops.txt
-bash: devops.txt: Permission denied
ubuntu@ip-172-31-44-9:~$
```

2. Remove execute permission from script

`chmod -x script.sh`

Try running:

`./script.sh`

Output:

Permission denied

```
ubuntu@ip-172-31-44-9:~$ chmod -x script.sh
ubuntu@ip-172-31-44-9:~$ ./script.sh
-bash: ./script.sh: Permission denied
ubuntu@ip-172-31-44-9:~$
```

What I Learned

- Linux permissions control file access for owner, group, and others.
- `chmod` is used to change permissions using symbols or numbers.
- Execute permission is required to run shell scripts.